

ENCODER PCB-R | FABRICATION DRAWING

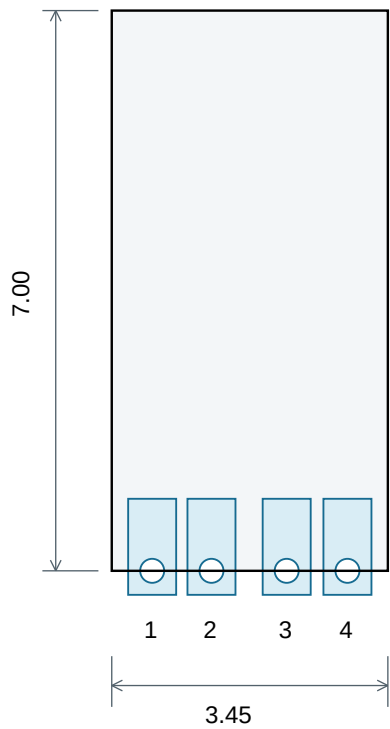
HM_Nightfall-mini-2e_Encoder-PCB-R_v1

CUSTOM PROCESS - MANUFACTURER REVIEW REQUIRED

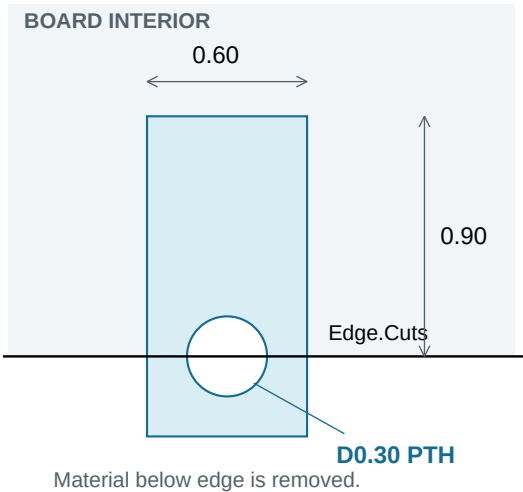
Units: mm | Nominal dimensions | Do not scale | 2026-09-21

Terminal layout (viewed from F.Cu)

Other pads / vias omitted in this view.



One terminal: copper / drill detail



Same land on F.Cu / B.Cu and F.Mask / B.Mask.
Full land: 0.60 x 1.20; center offset: 0.30 inward.
Retain plated half-barrel joining both faces.

Half-hole centers in KiCad coordinates (Gerber/Excellon Y is negative)

Terminal	Net	X (mm)	Y (mm)
EC_R1 / TP	+3V3	147.2811	108.5036
EC_R2 / TP	GND	148.0221	108.5036
EC_R3 / TP	EC_R_B	148.9619	108.5036
EC_R4 / TP	EC_R_A	149.7211	108.5036

Manufacturing notes

- Two copper layers, FR-4; nominal thickness 1.60 mm (as in source PCB).
- Four D0.30 plated half-holes; drill, plate, then route the straight edge through centers.
No NPTH substitution. Do not plate the whole edge or connect neighboring terminals.
- Side annular copper: 0.15; minimum hole edge gap: 0.4410.
- Also retain 6 existing D0.20 plated vias. NPTH count: 0. Silkscreen: none, both sides.
- Review small-board support/panelization, hole plating and finish before manufacture.
Keep tabs away from the terminal edge. Do not alter final board size or pad geometry.
- This design is outside standard small-castellation limits; custom acceptance is pending.

Existing D0.20 via drill sizes remain below the project 0.30 mm DRC minimum.